

```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv("bridges.csv")
```

```
In [3]: df.head()
```

```
Out[3]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>Id</th>
<th>river</th> <th>location</th> <th>erected</th>
<th>purpose</th> <th>length</th> <th>lanes</th>
<th>clear-g</th> <th>t-or-d</th> <th>material</th>
<th>span</th> <th>rel-l</th> <th>type</th>
</thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

|    |   |      |      |          |        |     |   |     |
|----|---|------|------|----------|--------|-----|---|-----|
| E1 | M | 3.0  | 1818 | HIGHWAY  | NaN    | 2.0 | N | THR |
| E2 | A | 25.0 | 1819 | HIGHWAY  | 1037.0 | 2.0 | N | THR |
| E3 | A | 39.0 | 1829 | AQUEDUCT | NaN    | 1.0 | N | THR |
| E5 | A | 29.0 | 1837 | HIGHWAY  | 1000.0 | 2.0 | N | THR |
| E6 | M | 23.0 | 1838 | HIGHWAY  | NaN    | 2.0 | N | THR |

```
In [4]: df.columns
```

```
Out[4]: Index(['Id', 'river', 'location', 'erecte
d', 'purpose', 'length', 'lanes',
               'clear-g', 't-or-d', 'material',
               'span', 'rel-l', 'type'],
              dtype='str')
```

```
In [5]: df["length"].describe()
```

```
Out[5]: count      81.000000  
mean    1567.469136  
std      747.491523  
min      804.000000  
25%     1000.000000  
50%     1300.000000  
75%     2000.000000  
max     4558.000000  
Name: length, dtype: float64
```

<p>I assume that the length shouldn't be externally short or long ,which mean I consider safety and convenience ,So i will flag the external long and short. which is (804.4558). I will screen the length column for values that appear externally short or externally long based on my chosen range.</p>

```
In [6]: flagged = df[(df["length"] < 870) | (df["length"] > 4558)]  
flagged
```

```
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<th>span</th> <th>rel-l</th> <th>type</th>
</thead> <tbody> <th>31</th> <th>44</th>
<th>52</th> <th>72</th> <th>80</th> <th>88</th>
<th>98</th> <th>104</th> </tbody> <tbody>
</tbody>
```

|       |   |      |      |         |    |        |     |   |   |
|-------|---|------|------|---------|----|--------|-----|---|---|
| E34   | O | 41.0 | 1888 |         | RR | 4558.0 | 2.0 | G | 7 |
| E46   | A | 37.0 | 1897 |         | RR | 4000.0 | 2.0 | G | 7 |
| E87   | A | 35.0 | 1903 |         | RR | 3000.0 | 2.0 | G | 7 |
| E70   | A | 27.0 | 1926 | HIGHWAY |    | 860.0  | 4.0 | G | 7 |
| E71   | A | 25.0 | 1928 | HIGHWAY |    | 860.0  | 4.0 | G | 7 |
| E107N | A | 39.7 | 1945 |         | RR | 840.0  | 2.0 | G | 7 |
| E82   | O | 42.0 | 1955 | HIGHWAY |    | 804.0  | NaN | G | 7 |
| E91   | O | 44.0 | 1975 | HIGHWAY |    | 3756.0 | 6.0 | G | 7 |

```
In [7]: flagged[["length"]].sort_values("length")
```

```
Out[7]: <style scoped> .dataframe tbody tr th:only-of-type {
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</thead> <tbody> <th>98</th> <th>88</th>
<th>72</th> <th>80</th> <th>52</th> <th>104</th>
<th>44</th> <th>31</th> </tbody> <tbody>
</tbody>
```

804.0

840.0

860.0

860.0

3000.0

3756.0

4000.0

4558.0

Handling the Flagged Values

I found 8 bridge lengths that were flagged which is extreme values. the range between( 804 to 4558). These values are extreme compared with the rest of the data, whoever, it is not impossible value . I will keep these rows in the dataset because I do not have enough evidence to say that the values are true .it might shown for anyother reason or constraction issue which I don't have details about it. I will document them as extreme values for further review.

In [ ]: